

LETTER • OPEN ACCESS

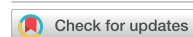
The effectiveness of heat prevention plans in reducing heat-related mortality across Europe

To cite this article: Aleš Urban *et al* 2025 *Environ. Res. Lett.* **20** 124071

View the [article online](#) for updates and enhancements.

You may also like

- [The one-stage design for nationwide studies on temperature and health](#)
Aurelio Tobias and Carmen Iñiguez
- [Quantifying the impact of changing the threshold of New York City heat emergency plan in reducing heat-related illnesses](#)
Tarik Benmarhnia, Lara Schwarz, Amruta Nori-Sarma et al.
- [The footprint of human-induced climate change on heat-related deaths in the summer of 2022 in Switzerland](#)
Ana M Vicedo-Cabrera, Evan de Schrijver, Dominik L Schumacher et al.

ENVIRONMENTAL RESEARCH
LETTERS

LETTER

OPEN ACCESS

RECEIVED

19 July 2025

REVISED

18 November 2025

ACCEPTED FOR PUBLICATION

3 December 2025

PUBLISHED

23 December 2025

Original content from
this work may be used
under the terms of the
[Creative Commons
Attribution 4.0 licence](#).

Any further distribution
of this work must
maintain attribution to
the author(s) and the title
of the work, journal
citation and DOI.

The effectiveness of heat prevention plans in reducing
heat-related mortality across Europe

Aleš Urban^{1,2,*} , Veronika Huber^{3,4} , Salomé Henry⁵, Nuria Pilar Plaza⁶ , Lucie Tušlová¹,
Shouro Dasgupta⁷ , Pierre Masselot⁸ , Ivana Cvijanovic⁹ , Malcolm Mistry^{8,10} , Mathilde Pascal¹¹,
Francesca de' Donato¹² , Claudia Di Napoli¹³ , Simon N Gosling¹⁴ , Silvia Kohnová¹⁵ , Jan Kyselý^{1,2},
Samuel Lüthi¹⁶ , Louis-François Pau^{17,18,19} , Martina S Ragetti^{20,21}, Reija Ruuhela²² , Niilo Rytö^{23,24},
Susana Das Neves Pereira da Silva²⁵ , Shiri Zemah-Shamir²⁶ , Wim Thiery²⁷,
Ana-Maria Vicedo-Cabrera^{28,29} , Joanna Wiecezorek³⁰ , Francesco Sera³¹, Ben Armstrong⁸ ,
Antonio Gasparrini⁸ , On behalf of the PROCLIAS TG 3.11
and the MCC Collaborative Research Network³²

¹ Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Prague, Czech Republic

² Institute of Atmospheric Physics, Czech Academy of Sciences, Prague, Czech Republic

³ Doñana Biological Station (EBD), Spanish National Research Council (CSIC), Sevilla, Spain

⁴ Institute of Epidemiology, Helmholtz Zentrum München—German Research Center for Environmental Health (GmbH), Neuherberg, Germany

⁵ AgroParisTech, Paris, France

⁶ Centro de Investigaciones sobre Desertificación, Consejo Superior de Investigaciones Científicas (CIDE, CSIC-UV-Generalitat Valenciana), Climate, Atmosphere and Ocean Laboratory (Climatoc-Lab), Moncada, Valencia, Spain

⁷ Centro Euro-Mediterraneo sui Cambiamenti Climatici (CMCC), Venezia, Italy

⁸ Environment & Health Modelling (EHM) Lab, Department of Public Health Environments and Society, London School of Hygiene & Tropical Medicine, London, United Kingdom

⁹ ESPACE-DEV, Univ Montpellier, IRD, Montpellier, France

¹⁰ Department of Economics, Ca' Foscari University of Venice, Venice, Italy

¹¹ Santé Publique France, Department of Environmental and Occupational Health, French National Public Health Agency, Saint Maurice, France

¹² Department of Epidemiology, Lazio Regional Health Service, ASL ROMA 1, Rome, Italy

¹³ European Centre for Medium-Range Weather Forecasts, Reading, United Kingdom

¹⁴ School of Geography, University of Nottingham, Nottingham, United Kingdom

¹⁵ Department of Land and Water Resources Management, Faculty of Civil Engineering, Slovak University of Technology, Bratislava, Slovakia

¹⁶ Weather and Climate Risks Group, Institute for Environmental Decisions, ETH Zürich, Switzerland

¹⁷ Rotterdam School of Management, Erasmus University Rotterdam, Rotterdam, The Netherlands

¹⁸ Copenhagen Business School, Copenhagen, Denmark

¹⁹ Uppgöta AB, Stockholm, Sweden

²⁰ Swiss Tropical and Public Health Institute, Allschwil, Switzerland

²¹ University of Basel, Basel, Switzerland

²² Weather and Climate Change Impact Research, Finnish Meteorological Institute, Helsinki, Finland

²³ Center for Environmental and Respiratory Health Research (CERH), Research Unit of Population Health, University of Oulu, Oulu, Finland

²⁴ Department of Public Health, University of Helsinki, Helsinki, Finland

²⁵ Department of Epidemiology, Instituto Nacional de Saúde Dr Ricardo Jorge, Lisbon, Portugal

²⁶ School of Sustainability, Reichman University, Herzliya, Israel

²⁷ Department of Water and Climate, Vrije Universiteit Brussel, Brussels, Belgium

²⁸ Institute of Social and Preventive Medicine, University of Bern, Bern, Switzerland

²⁹ Oeschger Centre for Climate Change Research, University of Bern, Bern, Switzerland

³⁰ Centre of Numerical Weather Prediction, Institute of Meteorology and Water Management—National Research Institute, Warsaw, Poland

³¹ Department of Statistics, Computer Science and Applications G. Parenti, University of Florence, Florence, Italy

³² See Acknowledgement.

* Author to whom any correspondence should be addressed.

E-mail: urban@ufa.cas.cz**Keywords:** heat prevention plans, heat and health warning systems, heat-related mortality, heat adaptationSupplementary material for this article is available [online](#)

Abstract

Heat-health warning systems and action plans, referred to as heat prevention plans (HPPs), are key public health interventions aimed at reducing heat-related mortality. Despite their importance, prior assessments of their effectiveness have yielded inconsistent results. The objective of this study is to systematically assess the effectiveness of HPPs in reducing heat-related mortality risk across Europe. We analysed daily mortality and mean temperature data from 102 locations in 14 European countries between 1990 and 2019. Using data from national experts, we identified the year of HPP implementation and categorised their development class. A three-stage analysis was conducted: (1) quasi-Poisson time series models were used to estimate location-specific warm-season exposure-response functions in 3 year subperiods; (2) mixed-effect meta-regression models with multilevel longitudinal structures were employed to quantify changes in pooled exposure-response functions due to HPP implementation, adjusted for long-term trends in heat-related mortality risks; and (3) the heat-related excess mortality due to HPP was calculated by comparing factual (with HPP) and counterfactual (without HPP) scenarios. Estimates are reported by country, region, and HPP class. HPP implementation was associated with a 25.2% [95% CI: 19.8% to 31.9%] reduction in excess deaths attributable to extreme heat, corresponding to 1.8 [95% CI: 1.3–2.4] avoided deaths annually per 100 000 inhabitants. This equates to an estimated 14 551 [95% CI: 10 118–19 072] total deaths avoided across all study locations following HPP implementation. No significant differences in HPP effectiveness were observed by European region or HPP class. Our findings provide robust evidence that HPPs substantially reduce heat-related mortality across Europe, accounting for temporal changes and geographical differences in risks. These results emphasise the importance of monitoring and evaluating HPPs to enhance adaptation to a warming climate.

1. Introduction

Europe is the fastest warming continent and has experienced an increase in extreme weather phenomena in the past two decades (EEA 2024). Intense heat waves occurred in South-Western Europe in 2003, in Eastern Europe and Russia in 2010, and across the whole continent in 2022 (Barriopedro *et al* 2011, Lhotka and Kyselý 2022, Ballester *et al* 2023, EEA-European Environment Agency 2024). Such events are likely to become a regular phenomenon in mid-latitudes by the end of the 21st century (CCAG 2024), and pose a high risk to human health.

Heat prevention plans (HPPs), including policies such as heat early warning systems and heat-health action plans, are universally considered key public health interventions to reduce health impacts of heat (Toloo *et al* 2013, McGregor *et al* 2015, Casanueva *et al* 2019). While widespread and adopted in many countries, evidence on the effectiveness of HPPs in reducing heat-related health risks is still limited. So far, most studies analysed the effect of HPP implementation in a single location by a simple comparison

of the heat-related mortality risk before and after the HPP implementation (Dwyer *et al* 2022). Only a few studies investigated beneficial effects of HPPs in multiple locations with inconsistent findings across cities and regions (De' Donato *et al* 2015, Martínez-Solanas and Basagaña 2019, Ragetti *et al* 2024). For example, a study comparing the temperature-mortality relationships in nine European cities before and after the major 2003 heatwave revealed reduced heat-related mortality risk in Rome and Paris, which implemented HPPs after 2003, but also in Athens, which did not (De' Donato *et al* 2015). In contrast, other cities that implemented HPPs after 2003 (e.g. Budapest, Valencia, London) did not see any significant reductions in heat-related mortality. Similarly, two studies from the US received inconclusive results in the beneficial effects of heat alerts issued by the US National Weather Service (Weinberger *et al* 2017, 2021).

A factor complicating the evaluation of HPP beneficial effects is the temporal and geographical variation in mortality risks due to the underlying changes in vulnerability (Boeckmann and Rohn 2014) and uneven occurrence of major heat wave

events across the continent (Lhotka and Kysely 2022). Trends in vulnerability may result from demographic changes, other policy interventions or autonomous adaptation to a changing climate. Previous studies have identified a long-term weakening of heat-attributable mortality risk in the past century (Sheridan and Allen 2018), despite warming trends in most regions of the world. However, the declining trend in the heat-attributable risk has slowed down or even reversed in most recent years, in spite of ongoing socioeconomic development and increasing implementation of HPPs (Pascal *et al* 2021, Urban *et al* 2022). In addition, there is limited knowledge of the comparative effectiveness of HPPs in different locations and with different level of development (Ebi 2019, Dwyer *et al* 2022). So far, only one study investigated the reduction effect of HPPs with accounting for the level of their development, revealing stronger mortality reductions in provinces with more advanced HPPs, though some regions with robust plans still reported increases in heat-attributable mortality (Martínez-Solanas and Basagaña 2019).

Despite using state-of-the-art methodology such as quasi-Poisson time series regression or difference-in-differences quasi experimental study design (Dwyer *et al* 2022), the main limitation of most studies is that they did not consider these underlying trends in heat-related mortality risks. The aim of this study is to provide a comprehensive assessment of the ability of HPPs to prevent heat-related mortality across Europe. The study benefits from the partnership between two consortia, the Multi-Country Multi-City (MCC) Collaborative Research Network and the COST Action PROCLIAS, allowing the collection of a large database including daily time series data for temperature and mortality as well as a detailed classification of HPPs in selected countries. The analysis is based on state-of-the-art epidemiological methods, with the application of a three-stage analysis using a multilevel and longitudinal design that allows separating effects of HPPs from underlying regional variations in mortality risks. This setting allows us to take into account a wide array of climatic conditions and long periods after the implementation of the HPPs.

2. Data and method

2.1. Mortality and temperature data

We obtained daily death counts and daily mean temperature data in European locations available via the MCC Network (<http://mccstudy.lshtm.ac.uk/>), an international collaboration of research teams producing epidemiological evidence on the association between environmental stressors, climate change, and health across the globe (Gasparrini *et al* 2024). For Europe, the MCC Network has gathered data from

ca. 280 locations in 19 countries. The current analysis was restricted to 102 locations in 14 European countries divided into four regions (i.e. four United Nations' M49 geographical subregions) with available data on HPPs implementation (figure 1(a)). Differently from the M49 classification, Belgrade, Serbia, was considered Eastern Europe for the purpose of this study due to its socioeconomic and climatological proximity to this region. Mortality data spanned the period from 1990 to 2019, whilst the exact timespans differed for individual countries (figure S1, in supplementary material). For each location, mortality is represented by daily counts for all causes, or for non-external causes (International Classification of Diseases (ICD-10): A0–R99). Daily mean temperature data from weather stations representative of the selected cities were collected by MCC partners from the respective national weather services (table S1).

2.2. HPPs

To assess the effectiveness of national policies to reduce heat-related mortality across Europe, we developed a catalogue of HPPs as part of the COST Action PROCLIAS in collaboration with national experts. The process relied on information collected in previously published studies (Casanueva *et al* 2019, Vanderplanken *et al* 2020, Dwyer *et al* 2022) as well as national documents available through the online portals Climate-ADAPT and Global Heat Health Information Network. The catalogue provides information for 34 European countries or regions, which were matched with the MCC locations (figure S2).

A HPP score was produced building upon the WMO/WHO's Guidance on warning-system development (table S2) (McGregor *et al* 2015) and the methodology proposed by (Martínez-Solanas and Basagaña 2019). Briefly, we defined a total score as the number of actions implemented within 8 core elements, assigning for each action included in the HPP points between 0 and 2 depending on their importance in preventing heat-related deaths. These weights were based on an expert elicitation event (<https://proclias.eu/news/elicitation-ws-prague-23>), where 22 international experts from the field of atmospheric sciences and/or public health evaluated the importance of individual actions. More details on the HPP score development is provided in Section S2 in the Supplementary Material.

2.3. Statistical analysis

All statistical analysis was conducted in the R software (version 4.4.0), using packages *dlnm*, *mixmeta*, and *lmtree*. The analysis was conducted using a three-stage multilevel longitudinal design to control for a spatial

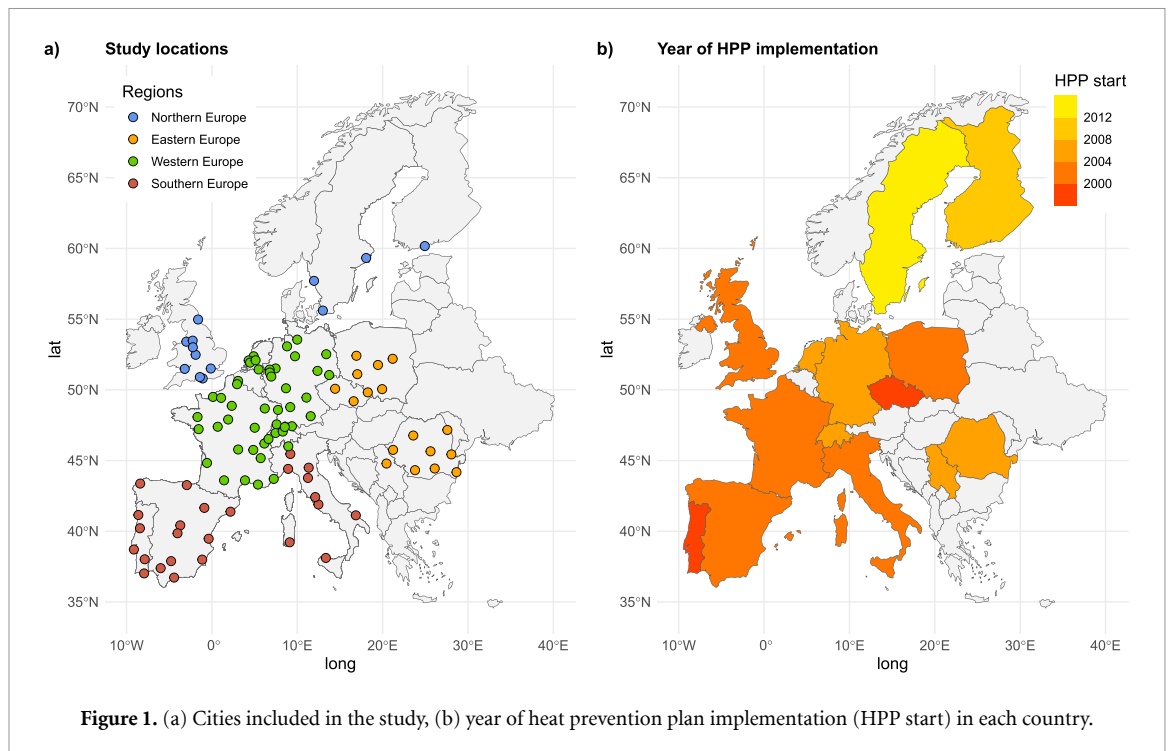


Figure 1. (a) Cities included in the study, (b) year of heat prevention plan implementation (HPP start) in each country.

and temporal confounding (Sera *et al* 2019, 2020). Each stage is described separately below.

Stage 1

First, we used a distributed lag nonlinear model (DLNM) with quasi-Poisson parametrisation (Gasparrini 2014), to estimate the temperature-mortality association in the five warmest months of the year (May–September, typically the season when HPPs are activated) for each location. The regression was fitted separately in three-year subperiods (from 1990–1992–2017–2019, figure S1). Daily mean temperature was chosen as the exposure variable, consistently with previous large-scale assessments (Masselet *et al* 2023). The model specification and parametrisation were based on previous studies focusing on the warm season (Gasparrini *et al* 2015, Sera *et al* 2020). Briefly, an interaction term between a natural cubic spline with four degrees of freedom for day of the season and year indicator was used to control for inter- and intra-seasonal variations in mortality data. In addition, we included a categorical variable for day of the week to control for a weekly cycle. The temperature-mortality relationship was modelled with a cross-basis function, using a quadratic B-spline with two internal knots placed at the 50th and 90th percentile of the full-period location-specific temperature distribution and a natural cubic spline with two internal knots equally distributed on a logarithmic scale over 10 d of lags to control for the lagged effects of heat and a short-term mortality displacement. The overall exposure-response function was then reduced into a one-dimensional overall cumulative exposure-response curve, which expresses

the location-specific relative risk (RR) of mortality compared to the subperiod- and location-specific minimum mortality temperature (MMT; Tobías *et al* 2021).

Stage 2

In the second stage, coefficients representing the reduced subperiod- and location-specific exposure-response functions were pooled in a multivariate mixed-effect meta-regression model with a multilevel longitudinal structure (Sera *et al* 2019). Six versions of the second-stage meta-regression model were compared to test specific aspects of heat-mortality associations and the effect of HPP implementation. Details of the models and their comparison are described in Section S3. The main model (Model 3) included i) an indicator for the presence of HPP, defined as 0 if its implementation occurred earlier or in the first year of the subperiod in which the HPP started, and 1 otherwise, and ii) an interaction term between Region (one of the four regions defined in figure 1) and (a) linear term (Time) for the middle of the subperiod, to control for region-specific long-term changes in heat-related mortality risks. The model included also random effects for intercept and time at city level, allowing differential baseline and trend in heat-related risk. Model 3 was further extended by using interaction terms with the HPP indicator to investigate differential effects of HPP implementation by Region (Model 4) and HPP score (Model 5), with the latter stratified in three groups (HPPclass). The estimation was carried out using a maximum-likelihood estimator and a diagonal structure of the random-effect (co)variance matrix. Statistical significance of

the fixed-effect interactions was tested by a likelihood ratio test (LRT) (Sera *et al* 2019).

Stage 3

In the third stage, we derived best linear unbiased predictions (BLUPs) from the meta-regression model to obtain improved location-specific estimates of exposure-response functions for each subperiod (Sera *et al* 2019). BLUPs were estimated by adding the residual random-effect part to fixed-effect predictions from the second-stage model, allowing them to be defined for two scenarios—*factual* and *counterfactual*. In the *factual* scenario, the HPP indicator corresponded to the observed pattern, while in the *counterfactual* scenario, the HPP indicator was set to 0 for all the subperiods. We then calculated the heat-attributable fraction of deaths (HAF) (Gasparrini and Leone 2014) in each location and each subperiod of the post-implementation period under the two scenarios. Since most HPPs have been designed to protect the population from extreme heat, we computed the HAF as the temperature-related percentage of deaths on days hotter than the 95th percentile of the location-specific summer temperature distribution over the whole study period. Finally, HAFs were used to calculate annual heat-attributable death rates per 100 000 inhabitants (HADs), using population estimates in each location for 2015 from EUROSTAT.

Eventually, the changes in HAFs (relative differences) and HADs (differences) between the counterfactual and factual scenarios in the post-implementation period were calculated to quantify the protective effect of HPP implementation in terms of the proportion of deaths avoided by HPP implementation in each country. To assess the impact of the 2003 heat wave on the results, we conducted the main analysis (Model 3) excluding data from the summer of 2003 as a sensitivity analysis.

The uncertainty of the HAF and HAD estimates, along with their changes, was quantified by generating 1000 samples of the BLUP coefficients using Monte Carlo simulations. The samples were generated assuming a multivariate normal distribution of coefficients of the second-stage meta-regression model (Massetot *et al* 2023). From these simulations, 95% empirical confidence intervals (CIs) were derived based on the empirical distribution of the coefficients.

3. Results

Out of 34 countries and regions listed in the HPP catalogue, data from 14 countries that provided suitable mortality as well as HPP data were included in the final analysis. The HPP scores based on the expert elicitation are shown in figure S2, with the UK and Portugal scoring the highest, having implemented actions across all core elements. Scandinavia (Sweden

and Finland) and central-Eastern Europe (Czechia and Poland), on the other hand, scored the lowest due to having heat alert systems with no follow-up actions (figure S3(a)). For further assessment, the HPPs were stratified in three levels (HPP class 1–3), defined as terciles of the maximum HPP score (table S3, figure S3(b)).

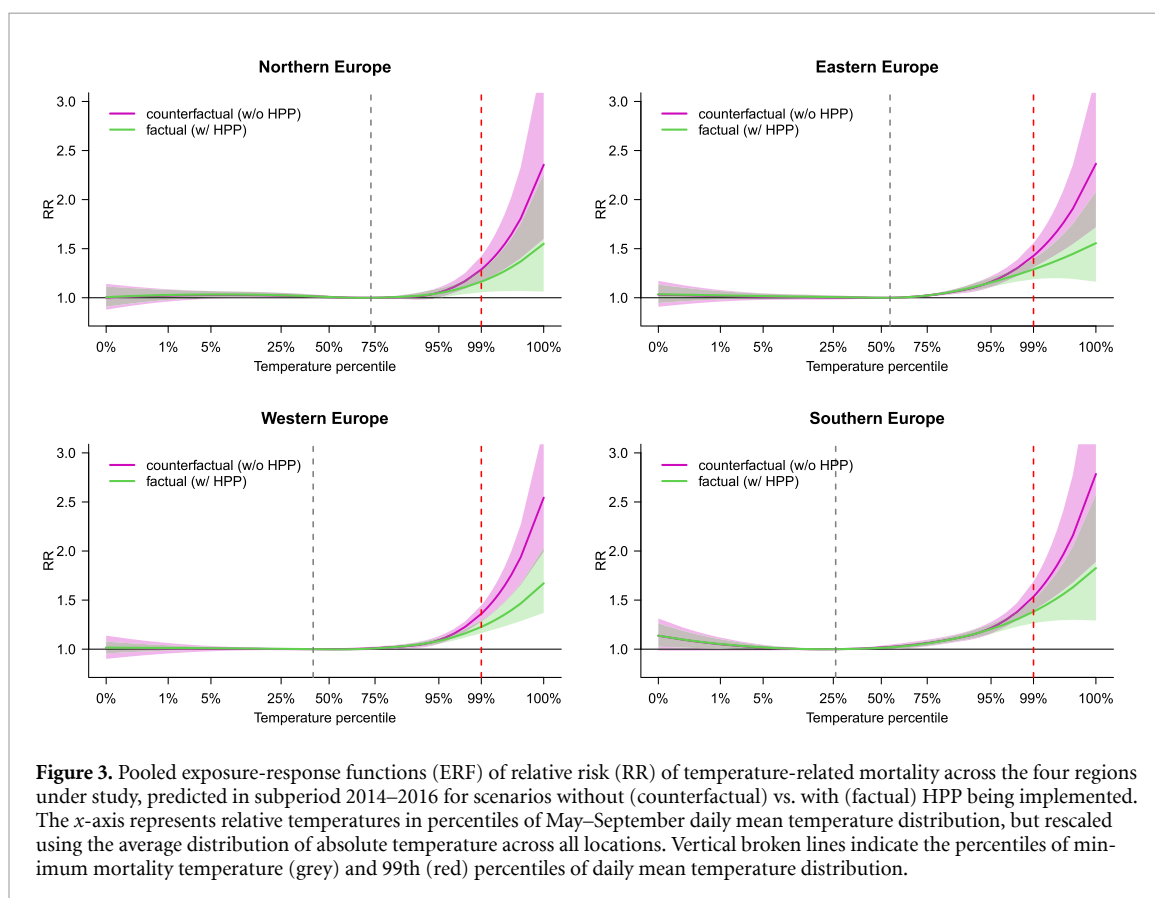
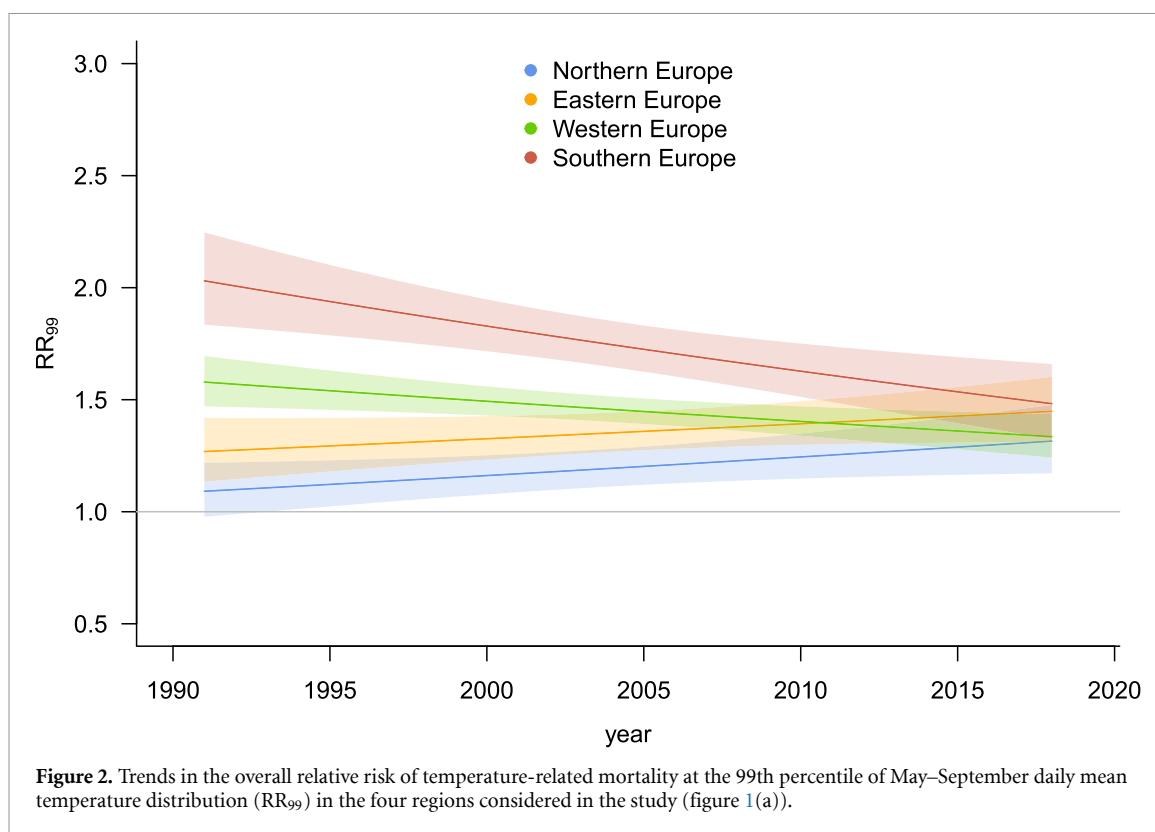
The year of implementation was the key element for analysing the impact of HPPs on heat-related mortality (figure 1(b)). Figure S1 shows that the most advanced HPPs (HPP class 3) were generally implemented in Southern and Western Europe after the 2003 heat wave, while the HPPs in Northern Europe (HPP class 1) were developed later as a response to the 2010 Eastern European heat wave. In general, less developed HPPs were found in Eastern and Northern European countries compared to Southern and Western Europe.

3.1. Trends in heat-related mortality

The comparison of the various specifications of the meta-analytical model indicates strong evidence for differential trends across regions (table S4). Therefore, the results from Model 3 are reported here. Figure 2 shows trends in the overall RR of temperature-related mortality at the 99th percentile of temperature distribution compared to the subperiod- and region-specific MMT (RR_{99}). The results indicate decreasing trends in RRs in populations of Southern (change in RR_{99} from 2.03 [95% CI: 1.84–2.25] in the first subperiod (1990–1992) to 1.48 [95% CI: 1.32–1.66] in the last one (2017–2019)) and Western Europe (1.58 [95% CI: 1.47–1.69] to 1.34 [95% CI: 1.24–1.44]). On the contrary, increasing trends in heat-related risks were observed in Northern (1.09 [95% CI: 0.98–1.22] to 1.32 [95% CI: 1.17–1.48]) and Eastern Europe (1.27 [95% CI: 1.13–1.42] to 1.45 [1.31–1.60]).

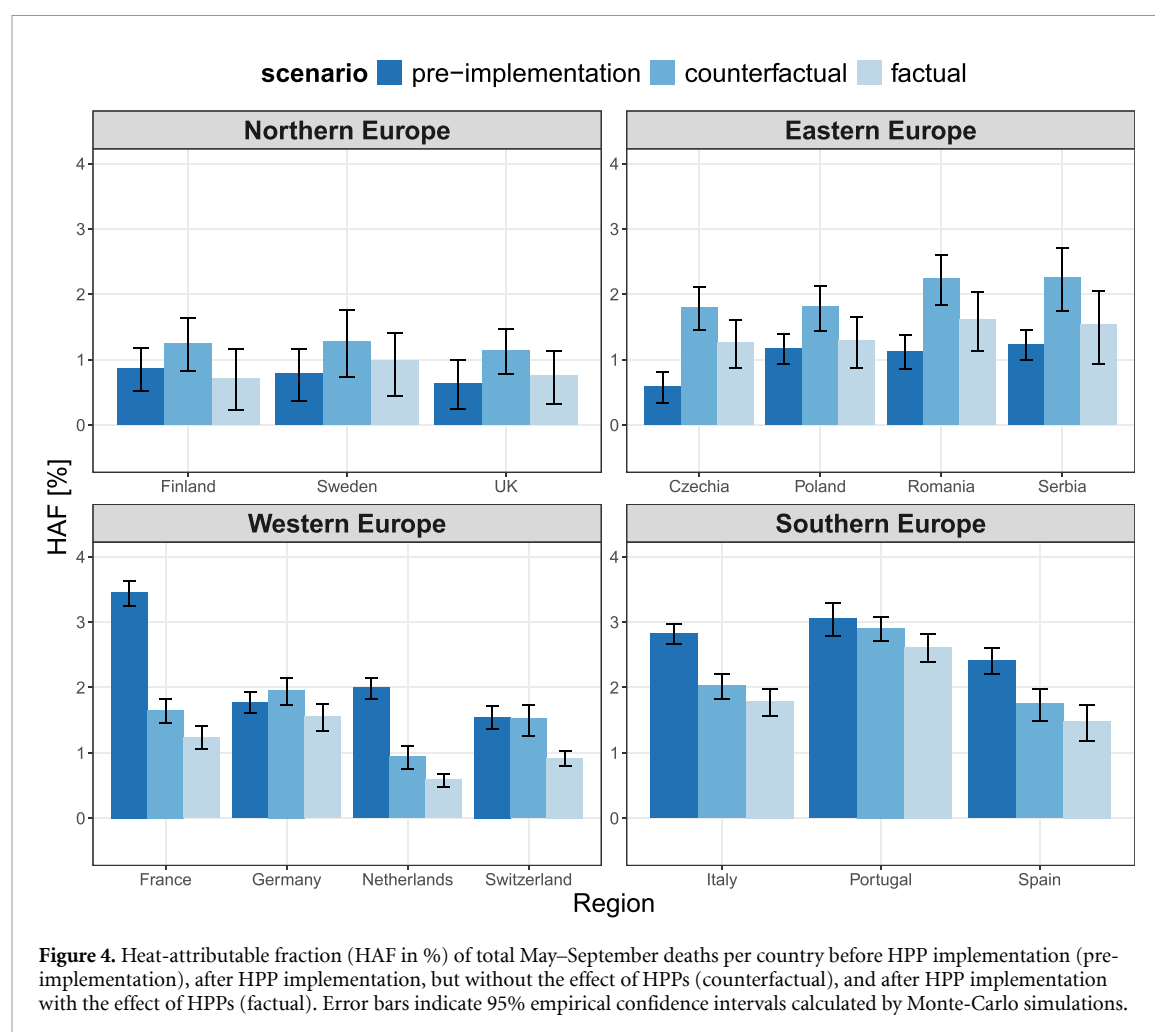
Figure 3 presents the difference in the pooled estimates of overall ERFs for the four regions predicted in 2015 under the two scenarios, namely without (*counterfactual*) and with (*factual*) HPP being implemented. These estimates are adjusted for regional temporal trends presented in figure 2, in addition to city-level deviations. Model 3 revealed strong evidence of an overall protective effect of HPP implementation according to LRT ($p < 0.001$) (table S4). On average, HPP implementation was associated with a reduction of RR_{99} from 1.41 [95% CI: 1.35–1.46] in the *counterfactual* to 1.27 [95% CI: 1.22–1.31] in the *factual* scenario (table S5). While the differences in RR_{99} were similar in all regions (in line with Model 3 settings), a comparison of the 95% CIs indicates that the reduction was most evident in Western Europe with $RR_{99} = 1.36$ [95% CI: 1.28–1.45] for the *counterfactual* and 1.23 [95% CI: 1.16; 1.29] for the *factual* scenario.

The analysis of differential effects of HPPs across regions (Model 4) seemed to suggest a larger decrease



in the heat-related mortality risk in Southern Europe, with RR_{99} of 1.61 [95% CI: 1.41–1.83] vs 1.39 [95% CI: 1.27–1.52] for the *counterfactual* and *factual* scenarios, respectively, compared to Eastern Europe (RR

of 1.37 [95% CI: 1.23–1.53] vs 1.29 [95% CI: 1.19–1.40]). However, LRT showed no statistical evidence ($p = 0.44$) of a differential effect of HPP based on Region (table S4).



Similarly, Model 5 revealed limited evidence (LRT $p = 0.19$) of differences in HPP implementation between the HPP classes (table S4). Indeed, while the model confirms the protective effect of any HPP implementation, small differences were found between the types of HPPs (table S6). Given the limited evidence of differential effects of Region and HPPclass in Models 4 and 5, Model 3 was used further to provide estimates of the changes in heat-attributable excess mortality with HPP implementation.

3.2. Effects of HPP implementation on heat-attributable excess mortality

Figure 4 shows variations in the mean HAF in individual countries in the period before HPP implementation (pre-implementation), compared to the period after HPP implementation under both the counterfactual and factual scenarios (see figure S4 for detailed temporal changes in HAFs over individual subperiods). Across all the locations considered in this study, implementation of HPPs was associated with a 25.2% [95% CI: 19.8–31.9%] decrease in HAF compared to the counterfactual scenario (table 1,

figure 5), which was equivalent to 1.8 [95% CI: 1.3–2.4] annual deaths per 100 000 inhabitants avoided due to HPPs. This implies that overall, 14 551 [95% CI: 10 118–19 072] deaths were prevented in locations of the study thanks to the implementation of HPPs.

Northern European countries were those with the largest impact of HPP on HAFs, with an average decrease in HAF of 33.2% [95% CI: 20.6–52.5%], while Southern Europe revealed the smallest reduction (HAF decrease of 11.9% [6.9–16.6%], table 1). However, the range of 95% CIs suggests a larger uncertainty of the estimates for Northern and Eastern Europe, likely related to the smaller number of locations considered in these regions (figure 5). This was true especially for Sweden with HAF decrease by 22.8% [95% CI: –30.8 to 68.6%] (an equivalent to annual 1.1 [95% CI: –2.1–4.9] avoided deaths per 100 000 inhabitants). In the UK, on the other hand, we observed the fourth largest and strong effect of HPP implementation according to mean HAF decrease (by 34.0% [95% CI: 19.5–55.8%]), although the change in HAD was the same as in Sweden (1.1 [95% CI: 0.6–1.6]). A sensitivity analysis revealed that even after excluding the summer of 2003 from

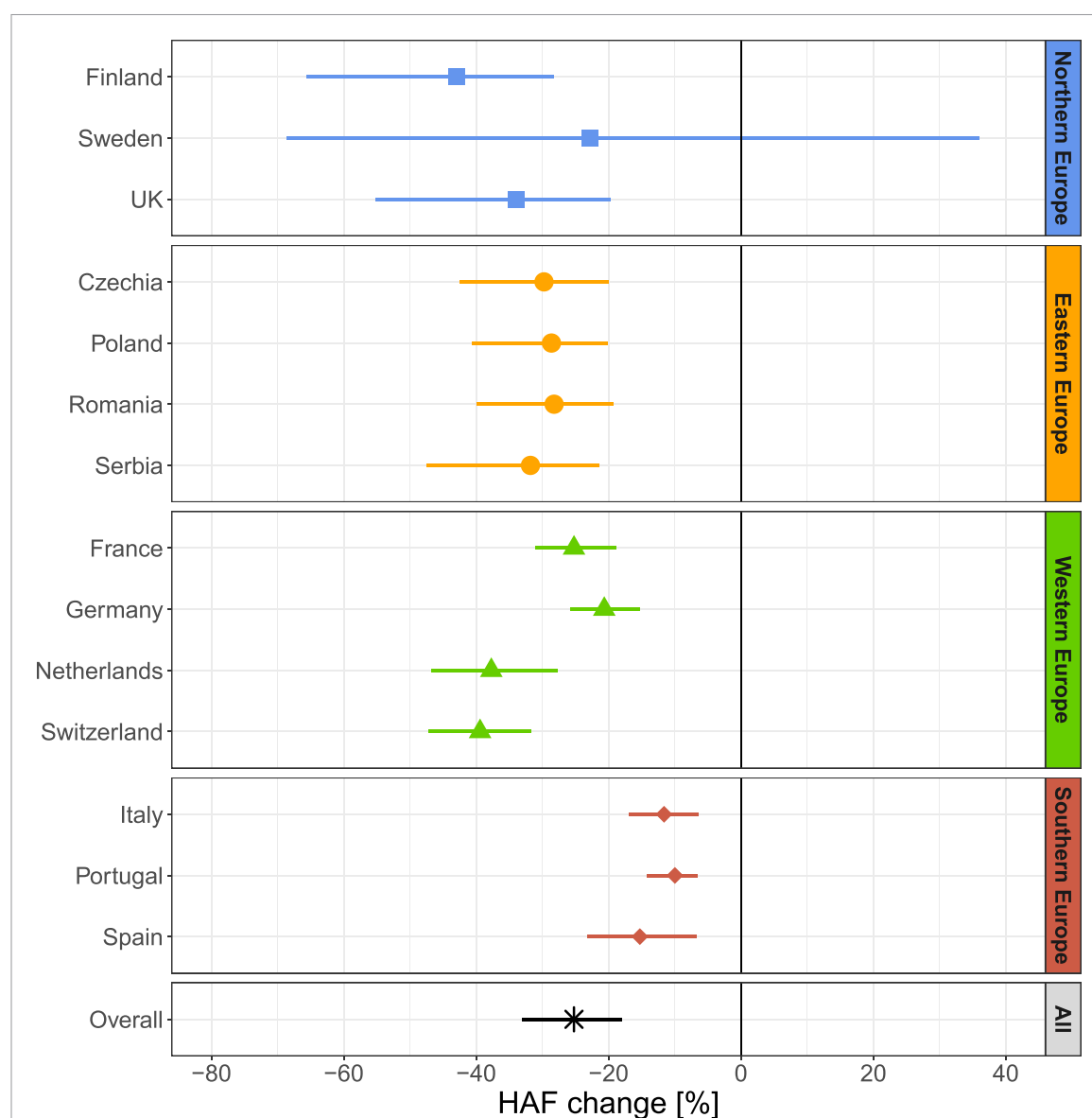


Figure 5. Change in heat-attributable mortality fraction (quantified as a relative difference between the *counterfactual* and the *factual* scenarios) per country associated with HPP implementation. Error bars indicate 95% empirical confidence intervals calculated by Monte-Carlo simulations (see table 1).

the analysis, the implementation of HPPs was associated with a weaker, but still evident reduction of heat-attributable mortality by 15.2% [95% CI: 4.1–23.7%] across all locations (figure S5).

4. Discussion

This study investigated the impact of HPP implementation on heat-related mortality across 102 locations in 14 European countries over 30 years. Results show that HPPs were associated with a 25% reduction in excess deaths attributable to extreme heat (HAF) compared to the counterfactual scenario without HPPs. This estimate accounted for differential temporal variations in heat-related mortality risks observed in the four European regions. Yet, counter to expectations, we found little evidence that

the protective effect of HPPs varied systematically by region or by HPP class.

4.1. Temporal changes in heat-related mortality

Studies have documented declining trends in heat-related mortality risk during past decades, attributed to positive socioeconomic development and health-care improvements (Boeckmann and Rohn 2014, Sheridan and Allen 2018). For instance, a global study of 305 locations in 10 developed countries revealed consistent reductions in heat-related mortality in all regions, except for Australia and Brazil (Vicedo-Cabrera *et al* 2018). However, more recent data suggest abatement of the declining trends or even increased heat-related mortality risks (Pascal *et al* 2021, Urban *et al* 2022, Ballester *et al* 2023). Our results also reflect this trend, especially in locations

Table 1. Relative differences in heat-attributable mortality fractions (HAF change) and absolute differences in heat-attributable deaths per 100 000 inh. (HAD difference) between the *counterfactual* and the *factual* scenarios in the study countries, regions and HPP classes. The values indicate the percentage of deaths and annual deaths per 100 000 inh., respectively, avoided by implementation of HPPs. The values are based on effect estimates of Model 3. Numbers in the brackets indicate 95% empirical confidence intervals estimated by Monte Carlo simulations.

Region	Country	HAF change (%)	HAD difference (annual deaths per 100 000 inh.)
Northern Europe	Finland	−43.0 (−66.6; −28.3)	−2.6 (−3.4; −1.9)
	Sweden	−22.8 (−68.6; 30.8)	−1.1 (−4.9; 2.1)
	UK	−34.0 (−55.8; −19.5)	−1.1 (−1.6; −0.6)
Eastern Europe	Czechia	−29.8 (−42.5; −20.6)	−2.3 (−2.9; −1.7)
	Poland	−28.6 (−41.3; −20.0)	−2.1 (−2.7; −1.5)
	Romania	−28.2 (−39.9; −19.8)	−2.2 (−2.8; −1.6)
	Serbia	−31.8 (−47.4; −21.5)	−4.3 (−5.5; −3.3)
Western Europe	France	−25.2 (−31.5; −19.1)	−1.4 (−1.7; −1.0)
	Germany	−20.6 (−25.7; −15.8)	−1.8 (−2.3; −1.4)
	Netherlands	−37.7 (−45.4; −29.6)	−1.1 (−1.5; −0.7)
	Switzerland	−39.4 (−47.2; −31.5)	−2.6 (−3.4; −1.9)
Southern Europe	Italy	−11.6 (−16.4; −6.7)	−0.9 (−1.2; −0.5)
	Portugal	−10.0 (−13.3; −6.7)	−1.2 (−1.6; −0.8)
	Spain	−15.3 (−23.0; −7.2)	−0.8 (−1.2; −0.4)
Mean per region	Northern Europe	−33.2 (−52.5; −20.7)	−1.6 (−2.9; −0.5)
	Eastern Europe	−29.6 (−42.8; −20.7)	−2.8 (−3.5; −2.0)
	Western Europe	−29.2 (−34.8; −23.7)	−1.7 (−2.2; −1.3)
	Southern Europe	−11.9 (−16.6; −6.9)	−1.0 (−1.3; −0.6)
Overall mean		−25.2 (−31.9; −19.8)	−1.8 (−2.4; −1.3)

where the most recent mortality data were available (i.e. for the subperiod 2017–2019). This suggests that accelerated trends in increasing frequency and intensity of heatwaves may challenge heat prevention and adaptation measures in the whole of Europe (Pascal *et al* 2021, Miranda *et al* 2023).

4.2. The effect of HPP implementation

Previous research suggested that HPPs may reduce adverse health outcomes associated with heat, although in some cases using indirect evidence and with inconsistent findings. The epidemiological literature suggests that other factors, such as climate variability, demographic changes, physiological adaptation, healthcare system improvements, and general socio-economic development (Boeckmann and Rohn 2014), play an important role and can generate temporal trends in heat vulnerability. Despite the recognised importance of these factors, most previous assessments of the effectiveness of HPPs have not adjusted for underlying trends in heat vulnerability, nor have they considered temporal variations in temperature distribution. The longitudinal meta-regression used in this study enabled us to adjust the impact of HPP implementation for both overall trends in heat-related mortality related to long-term changes in mortality data (e.g. population aging) and spatio-temporal variations in temperature distribution (e.g. occurrence of major heat wave

events), and to reveal a beneficial effect of HPP implementation even in countries with increasing heat vulnerability.

4.3. The role of HPP class

To our knowledge, no study to date has provided a comprehensive analysis of the beneficial effect of HPPs across Europe, taking into account different characteristics of HPPs in individual countries and separating the effect of HPP implementation from the underlying trends. A nation-wide study in Spain found stronger mortality reductions in provinces with more advanced HPPs, though some regions with robust plans still reported increases in heat-attributable mortality (Martínez-Solanas and Basagaña 2019). Our results revealed limited evidence that more developed HPPs result in a stronger protective effect. This may reflect large uncertainties associated with the HPP data collection and evaluation. In each country, the effective operation of individual HPP elements is subject to actual implementation at the regional and local scale, which is influenced by various factors, such as the amount of financial and human resources provided for HPP implementation, the level of harmonization between national and regional action plans, or the level of involvement of local health and social institutions (Martínez *et al* 2022). Although several studies evaluated the status of HPPs in Europe (Casanueva *et al* 2019, Vanderplanken *et al* 2020, Dwyer *et al*

2022), none of them were able to collect information fully representative for the whole of Europe and all regional levels due to difficulties with data collection (Martinez *et al* 2022). Similarly, in this study we were not able to compare the HPPs at the regional and local level and verify if the actions declared at the national level were really implemented in each location. Therefore, the national HPP score may not fully reflect the actual situation in each location.

Moreover, different political and socio-economic conditions may influence the effectiveness of similar HPP elements across countries, leading to variability in estimated HAF decreases even among locations with comparable HPP scores. As summers similar to or even hotter than those of 2003, 2010 or 2022 are expected to become more frequent in Europe in the next decade, it is essential to intensify cross-border exchange of the good practice in development and implementation of HPPs, including a standardised evaluation of national and regional HPPs and a larger focus on long-term adaptation measures (Pascal *et al* 2021, Martinez *et al* 2022).

4.4. Limitations

Beside the challenges in HPP data collections, several limitations of the study need to be acknowledged. First, the analysis included data from a selection of cities in selected countries, excluding non-urban areas, that may not be fully representative of the four European regions. Second, temporal variations in heat-related mortality risks are modelled with region-specific linear terms, which can fail to capture more complex and local trends. Third, the analysis does not directly control for seasonal behavioural factors such as mass European-wide tourism during the summer season. Given the character of mortality data, which is defined by the place of residence of each deceased, it is impossible to track these patterns. In addition, the analysis does not address local environmental quality status (e.g. air pollution) as well as it did not directly account for specific drivers of heat adaptation such as air conditioning, which has been previously found as associated with a reduction in heat-related risks (Sera *et al* 2020), although it is expected that its effects are incorporated into the temporal trends.

5. Conclusion

This study provides the most comprehensive assessment to date of the potential of HPPs for reducing heat-related mortality in Europe. It systematically explores the association between HPP implementation and heat-related mortality, employing state-of-the-art epidemiological designs and statistical techniques that account for temporal variations and geographical differences in heat-related mortality. This

robust evaluation highlights a significant association between the implementation of HPPs and a 25% reduction in heat-related mortality across 102 locations in 14 European countries.

While our findings indicate that HPPs are generally effective in reducing heat-related excess mortality, challenges include limited access to accurate information on HPPs in specific locations, inconsistent regional implementation, and difficulties in tracking HPP updates. Given the increasing frequency and intensity of heatwaves in Europe, the findings emphasise the importance of a nuanced understanding of broader determinants of heat vulnerability. Refining and unifying HPP evaluation metrics is crucial to ensure a comprehensive assessment of their effectiveness in preventing heat-related mortality across Europe.

Contributors

AU, VH, SD, SNG, FS, BA, PM, AG were involved in the design of the study. SH, NPP, and LT contributed to the HPP data collection and evaluation. AU and VH accessed and verified the data provided by the co-authors. AU performed the analyses, drafted an internal summary report to circulate preliminary results and figures, and wrote the first version of the manuscript. AU, VH, FS, BA, PM, and AG contributed to the methodological framework. All authors had full access to all the data in the study, contributed to the interpretation of the results, reviewed and edited the manuscript, and had final responsibility for the decision to submit for publication.

Data availability statement

Anonymised mortality data used in this study were collected by collaborators within the MCC Network under a data-sharing agreement (<http://mccstudy.lshtm.ac.uk/>) and cannot be made publicly available. The exposure-response curve coefficients from the Stage 1 analysis, along with the complete R code for replicating the subsequent analyses, will be made available in the GitHub (<https://github.com/urbanales/MCC-HEWS>) and Zenodo (<https://doi.org/10.5281/zenodo.17507283>) repositories within one month of manuscript acceptance. Researchers interested in accessing the raw mortality data for the included cities may contact the corresponding author (Aleš Urban, urban@ufa.cas.cz) or the MCC Network coordinators.

Acknowledgments

We would like to thank all experts within the COST Action 'PROCLIAS' (<https://proclias.eu/>) and the MCC Collaborative Research

Network (<http://mccstudy.lshtm.ac.uk/>) who helped us with the collection of HPP data. Additionally, we thank 22 experts who participated in the expert elicitation workshop and provided their expertise on the HPPs and their core elements.

Following members of the MCC Collaborative Research Network contributed to the mortality data collection.

Jouni J K Jaakkola^{1,2}, Alexandra Schneider³, Paola Michelozzi⁴, Jochem Klompmaker^{5,6}, Joana Madureira^{7,8,9}, Iulian-Horia Holobaca¹⁰, Aurelio Tobias¹¹, Carmen Íñiguez^{12,13}, Bertil Forsberg¹⁴, Daniel Rabczenko¹⁵.

1. Center for Environmental and Respiratory Health Research (CERH), Research Unit of Population Health and Biocenter Oulu, University of Oulu, Oulu, Republic of Finland; 2. Medical Research Center Oulu, Oulu University Hospital and University of Oulu, Oulu, Republic of Finland; 3. Institute of Epidemiology, Helmholtz Zentrum München—German Research Center for Environmental Health (GmbH), Neuherberg, Germany, Bertil Forsberg; 4. Department of Epidemiology, Lazio Regional Health Service, Rome, Italian Republic; 5. Centre for Sustainability, Environment and Health, National Institute for Public Health and the Environment, Bilthoven, Kingdom of the Netherlands, Netherlands; 6. Institute for Risk Assessment Sciences, Utrecht University, Utrecht, Kingdom of the Netherlands, Netherlands; 7. Department of Environmental Health, Instituto Nacional de Saúde Dr Ricardo Jorge, Porto, Portuguese Republic; 8. EPIUnit – Instituto de Saúde Pública, Universidade do Porto, Porto, Portuguese Republic; 9. Laboratório para a Investigação Integrativa e Translacional em Saúde Populacional (ITR), Porto, Portuguese Republic; 10. Faculty of Geography, Babes–Bolyai University, Cluj–Napoca, Romania; 11. Institute of Environmental Assessment and Water Research (IDAEA), Spanish Council for Scientific Research (CSIC), Barcelona, Kingdom of Spain; 12. Department of Statistics and Computational Research, Universitat de València, València, Kingdom of Spain; 13. Epidemiology and Public Health Networking Biomedical Research Centre (CIBERESP), Madrid, Kingdom of Spain; 14. Department of Public Health and Clinical Medicine, Umeå University, Kingdom of Sweden; 15. National Institute of Hygiene, Warszawa, Republic of Poland.

Conflict of interest

We declare no competing interests.

Ethical approval

Not required. Anonymised aggregated data were used.

Funding

The collection of HPP data and the expert elicitation workshop were supported by COST Action ‘PROCLIAS’ (Grant ID: 19139). AU and JK were supported by the Czech Science Foundation (Grant ID: 22–24920S). AU acknowledges funding from the European Union’s Horizon Europe project ‘CROSSEU’ (Grant ID: 101081377) and the Czech Ministry of Education Youth and Sport’s programme ERC CZ (Grant ID: LL2410). VH was supported by the EU’s Horizon 2020 research and innovation programme (Marie Skłodowska–Curie Grant Agreement No.: 101032087). AMVC acknowledges funding from the Swiss National Science Foundation (Grant ID: TMSGI3_211626). WT received funding from the European Research Council (ERC) under the European Union’s Horizon Framework research and innovation programme (Grant ID: 101076909; ‘LACRIMA’ project). AG acknowledges funding from Medical Research Council–UK (Grant ID: MR/V034162/1), European Union’s Horizon 2020 Project ‘Exhaustion’ (Grant ID: 820655) and Wellcome Trust (Grant ID: 308914/Z/23/Z).

ORCID iDs

Aleš Urban  0000-0003-4608-3553
 Veronika Huber  0000-0001-9633-2752
 Nuria Pilar Plaza  0009-0004-5395-7858
 Shouro Dasgupta  0000-0003-4080-8066
 Pierre Masselot  0000-0002-7326-1290
 Ivana Cvijanovic  0000-0002-1738-7745
 Malcolm Mistry  0000-0003-3345-6197
 Francesca de’Donato  0000-0002-2225-9457
 Claudia Di Napoli  0000-0002-4901-3641
 Simon N Gosling  0000-0001-5973-6862
 Silvia Kohnová  0000-0002-5963-3748
 Samuel Lüthi  0000-0003-2884-3467
 Louis-François Pau  0000-0003-1503-1486
 Reija Ruuhela  0000-0001-7837-3366
 Susana Das Neves Pereira da Silva  0000-0003-2524-0548
 Shiri Zemah-Shamir  0000-0003-0651-0661
 Ana-Maria Vicedo-Cabrera  0000-0001-6982-8867
 Joanna Wiecezorek  0000-0001-6472-257X
 Ben Armstrong  0000-0003-4407-0409
 Antonio Gasparrini  0000-0002-2271-3568

References

- Ballester J, Quijal-Zamorano M, Méndez Turrubiates R, Pegenaute F, Herrmann F, Robine J M, Basagaña X, Tonne C, Antó J M and Achebak H 2023 Heat-related mortality in Europe during the summer of 2022 *Nat. Med.* **29** 1857–66
- Barriopedro D, Fischer E M, Luterbacher J, Trigo R M and García-Herrera R 2011 The hot summer of 2010: redrawing the temperature record map of Europe *Science* **332** 220–4
- Boeckmann M and Rohn I 2014 Is planned adaptation to heat reducing heat-related mortality and illness? A systematic review *BMC Public Health* **14** 1112

- Casanueva A, Burgstall A, Kotlarski S, Messeri A, Morabito M, Flouris A D, Nybo L, Spirig C and Schwierz C 2019 Overview of existing heat–health warning systems in Europe *Int. J. Environ. Res. Public Health* **16** 2657
- CCAG – Climate Crisis Advisory Group 2024 Record–breaking heatwave will be an average summer by 2035, latest met office Hadley centre data shows online (available at: www.preventionweb.net/news/record-breaking-heatwave-will-be-average-summer-2035-latest-met-office-hadley-centre-data)
- De' Donato F K *et al* 2015 Changes in the effect of heat on mortality in the last 20 years in nine European cities. Results from the PHASE project *Int. J. Environ. Res. Public Health* **12** 15567–83
- Dwyer I J, Barry S J E, Megiddo I and White C J 2022 Evaluations of heat action plans for reducing the health impacts of extreme heat: methodological developments (2012–2021) and remaining challenges *Int. J. Biometeorol.* **66** 1915–27
- Ebi K L 2019 Effective heat action plans: research to interventions *Environ. Res. Lett.* **14** 10–12
- EEA – European Environment Agency 2024 European climate risk assessment (available at: www.eea.europa.eu/en/about/who-we-are/projects-and-cooperation/agreements/european-climate-risk-assessment)
- Gasparrini A 2014 Modeling exposure–lag–response associations with distributed lag non–linear models *Stat. Med.* **33** 881–99
- Gasparrini A *et al* 2015 Temporal variation in heat–mortality associations: a multicountry study *Environ. Health Perspect.* **123** 1200–7
- Gasparrini A and Leone M 2014 Attributable risk from distributed lag models *BMC Med. Res. Methodol.* **14** 55
- Gasparrini A, Vicedo-Cabrera A M and Tobias A 2024 The multi–country multi–city collaborative research network an international research consortium investigating environment, climate, and health *Environ. Epidemiol.* **8** e339
- Lhotka O and Kyselý J 2022 The 2021 European heat wave in the context of past major heat waves *Earth Sp. Sci.* **9** 1–12
- Martinez G S, Kendrovski V, Salazar M A, de'Donato F and Boeckmann M 2022 Heat–health action planning in the WHO European Region: status and policy implications *Environ. Res.* **214** 113709
- Martínez-Solanas E and Basagaña X 2019 Temporal changes in temperature–related mortality in Spain and effect of the implementation of a heat health prevention plan *Environ. Res.* **169** 102–13
- Masselot P *et al* 2023 Excess mortality attributed to heat and cold: a health impact assessment study in 854 cities in Europe *Lancet Planet Heal* **7** e271–81
- McGregor G R, Bessemoulin P, Ebi K and Menne B 2015 Heatwaves and health: guidance on warning–system development (available at: www.who.int/globalchange/publications/WMO_WHO_Heat_Health_Guidance_2015.pdf)
- Miranda N D, Lizana J, Sparrow S N, Zachau-Walker M, Watson P A G, Wallom D C H, Khosla R and McCulloch M 2023 Change in cooling degree days with global mean temperature rise increasing from 1.5 °C to 2.0 °C *Nat. Sustain.* **6** 1326–30
- Pascal M, Lagarrigue R, Tabai A, Bonmarin I, Camail S, Laaidi K, Le Tertre A and Denys S 2021 Evolving heat waves characteristics challenge heat warning systems and prevention plans *Int. J. Biometeorol.* **65** 1683–94
- Ragettli M S, Flückiger B, Vienneau D, Domingo-Irigoyen S, Koschenz M and Rössli M 2024 Vulnerability to heat–related mortality and the effect of prevention measures: a time–stratified case–crossover study in Switzerland *Swiss Med. Wkly.* **154** 3410
- Sera F *et al* 2020 Air conditioning and heat–related mortality: a multi–country longitudinal study *Epidemiology* **31** 779–87
- Sera F, Armstrong B, Blangiardo M and Gasparrini A 2019 An extended mixed-effects framework for meta-analysis *Stat. Med.* **38** 5429–44
- Sheridan S C and Allen M J 2018 Temporal trends in human vulnerability to excessive heat *Environ. Res. Lett.* **13** 043001
- Tobías A *et al* 2021 Geographical variations of the minimum mortality temperature at a global scale *Environ. Epidemiol.* **5** e169
- Toloo G, Fitzgerald G, Aitken P, Verrall K and Tong S 2013 Are heat warning systems effective? *Environ. Health Glob. Access Sci. Source* **12** 27
- Urban A, Fonseca-Rodríguez O, Di Napoli C and Plavcová E 2022 Temporal changes of heat–attributable mortality in Prague, Czech Republic, over 1982–2019 *Urban Clim.* **44** 101197
- Vanderplanken K, van Loenhout J, Inac Y, Guha–Sapir D, van den Hazel P, Louis V, Shams A and Marx M 2020 *Critical analysis of heat plans and interviews. EU Project SCORCH. Deliverable 2.4* (available at: <https://heathealth.info/wp-content/uploads/D-2.4-Critical-Analysis-of-Heat-Plans-and-Interviews.pdf>)
- Vicedo-Cabrera A M *et al* 2018 A multi–country analysis on potential adaptive mechanisms to cold and heat in a changing climate *Environ. Int.* **111** 239–46
- Weinberger K R *et al* 2021 Heat warnings, mortality, and hospital admissions among older adults in the United States *Environ. Int.* **157** 106834
- Weinberger K R, Haykin L, Eliot M N, Schwartz J D, Gasparrini A and Wellenius G A 2017 Projected temperature–related deaths in ten large U.S. metropolitan areas under different climate change scenarios *Environ. Int.* **107** 196–204